UniTrade - NFR Evidence Pack

Project: UniTrade | Commit: 327b848c31f8 | Generated (UTC): 2026-10-26 08:57:50 UTC
21 requirements across 7 SRS quality attributes | pass: 21
Validation: PASS - all evidence references resolve
Evidence labels are clickable (committed files link to the commit permalink; CI items link to the test file and the workflow).

Administration

ID	Requirement	Target	Tool	Status	Evidence
QR-07a	Moderation queue returns cases oldest-first	oldest-first ordering	Integration test	pass	test: AdminSpineTests.cs CI: Backend CI
QR-07b	Every admin decision is audited	actor + timestamp + reason present (exactly one row)	Integration test	pass	test: AdminSpineTests.cs CI: Backend CI
QR-07c	Decision notifies the subject	Notification dispatched on each decision	Integration test (notification check)	pass	test: AdminDecisionTests.cs CI: Backend CI
QR-07d	SLA breach is surfaced on the dashboard	Cases older than SLA (48 h verification / 72 h disputes) flagged	Integration test	pass	test: AdminDecisionTests.cs CI: Backend CI

Maintainability

ID	Requirement	Target	Tool	Status	Evidence
QR-05a	CI build & test gate on main	Zero compile errors, 100% test pass rate and formatting enforced before merge	dotnet build / dotnet format / xUnit; tsc / ESLint / Vitest; SonarCloud static analysis	pass	CI: Backend CI CI: Web Frontend CI
QR-05b	Migrations apply cleanly from an empty database	Applies clean on a fresh Postgres	dotnet ef database update	pass	CI: Backend CI

Performance

ID	Requirement	Target	Tool	Status	Evidence
QR-01a	p95 read latency (feed / search / browse) under load	p95 <= 800 ms, error rate < 1% at ~40 concurrent users (baseline)	k6	pass	perf-summary(QR-01a).json
QR-01b	Chat message delivery latency	p95 <= 500 ms	Playwright (SignalR round-trip timing)	pass	chat-latency.json

Reliability

ID	Requirement	Target	Tool	Status	Evidence
QR-04a	Duplicate payment notification is idempotent	Exactly one side effect (one transaction, status completed)	Integration + duplicate delivery	pass	test: PaymentWebhookTests.cs CI: Backend CI
QR-04b	A single failed worker cycle does not stop the worker	Worker continues to next cycle after an injected fault	Integration + fault injection	pass	test: ReservationExpiryWorkerTests.cs CI: Backend CI
QR-04c	A failed deployment auto-rolls-back	Health-gated: health check before traffic shift; automatic rollback on failure	Deploy config + observed failing run	pass	cd.yml QR-04c_ Failed deployment should automatically roll back to the last live revision.pdf
QR-04d	Availability over a rolling 30-day window	>= 99.5%	UptimeRobot	pass	UpTimeRobotEvidence.png

Scalability

ID	Requirement	Target	Tool	Status	Evidence
QR-02	Error rate + latency at 2x baseline load	p95(2x) <= 1.5x baseline p95, error < 1% (baseline 40 VUs -> peak 80 VUs)	k6	pass	scale-summary(QR-02).json

Security

ID	Requirement	Target	Tool	Status	Evidence
QR-03a	Admin endpoints reject non-admin callers	100% reject (anon -> 401, student -> 403, admin -> 200)	Integration test	pass	test: AdminSpineTests.cs CI: Backend CI
QR-03b	Forged/tampered payment callback has no effect	Rejected (400 invalid_signature), no state change	Integration + fault injection	pass	test: PaymentWebhookTests.cs CI: Backend CI
QR-03c	OTP storage + comparison	Peppered hash; constant-time compare; never equals raw OTP	Unit test	pass	test: OtpSecurityTests.cs CI: Backend CI
QR-03d	No medium-or-above exposure on staging (ZAP baseline)	0 high/critical; mediums triaged	OWASP ZAP baseline	pass	zap-report(backend-QR-03d).html zap-report(backend-QR-03d).json zap-report(frontend-QR-03d).html zap-report(frontend-QR-03d).json
QR-03e	No high/critical dependency CVEs on main	0 high/critical	dotnet list package --vulnerable + npm audit --audit-level=high	pass	npm-audit-report.txt dotnet-vulnerability-report.txt
QR-03f	No high/critical OS or library CVEs in container images	0 high/critical (triaged entries listed in .trivyignore)	Trivy (container image scan)	pass	trivy-backend.txt trivy-frontend.txt

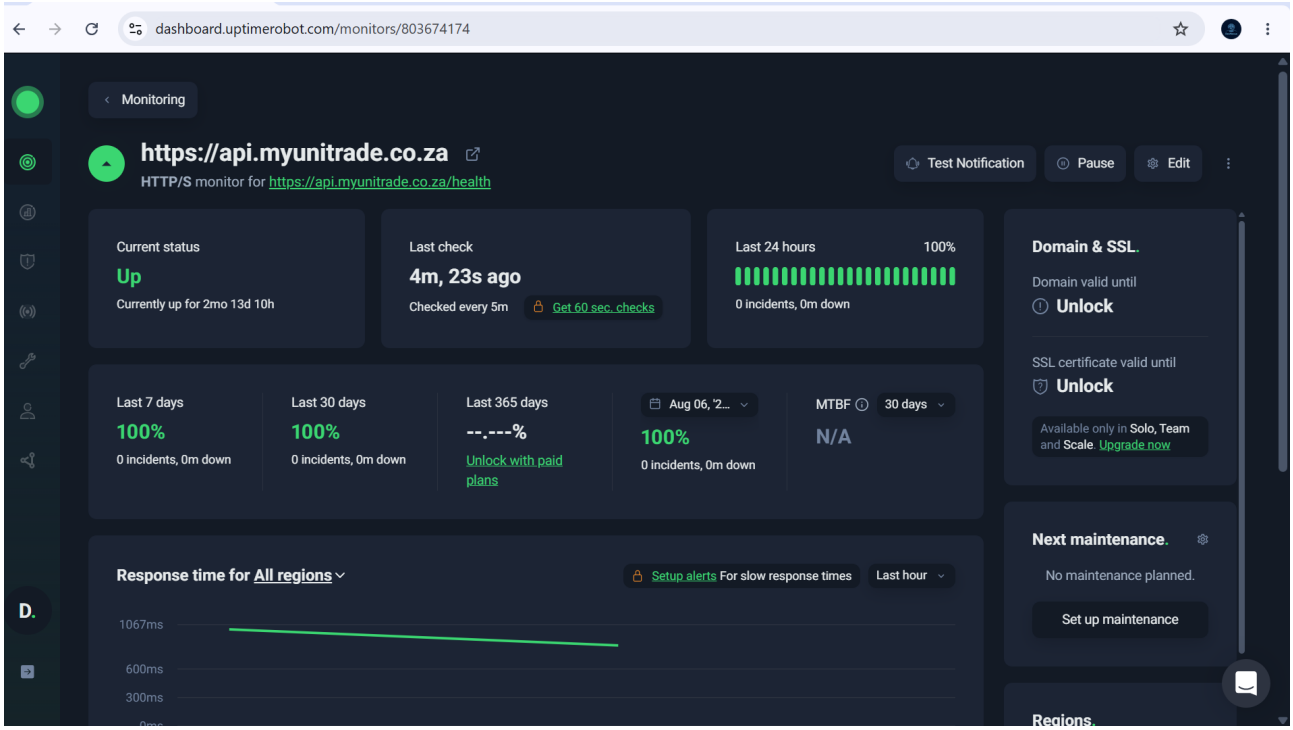
Usability

ID	Requirement	Target	Tool	Status	Evidence
QR-06a	Accessibility on public + admin pages	0 serious/critical (Lighthouse/axe >= 90 threshold)	Lighthouse (axe-core)	pass	perf-access-lighthouse/ (9 files)
QR-06b	Core task completes end-to-end in <= 2 min	<= 120s {browse -> reserve -> chat -> meetup -> initiate payment}	Playwright e2e	pass	core-task.json

Key measurements

ID	Metric	Measured	Target
QR-01a	p95 read latency	**43.8 ms**	p95 <= 800 ms, error rate < 1% at ~40 concurrent users (baseline)
QR-01a	request error rate	**0.00%**	p95 <= 800 ms, error rate < 1% at ~40 concurrent users (baseline)
QR-02	baseline p95	**51.4 ms**	p95(2x) <= 1.5x baseline p95, error < 1% (baseline 40 VUs -> peak 80 VUs)
QR-02	peak p95 (2x load)	**65.0 ms**	p95(2x) <= 1.5x baseline p95, error < 1% (baseline 40 VUs -> peak 80 VUs)

Availability evidence (QR-04d)



Integrity hashes for every committed evidence file are in tests/nfr/evidence/SHA256SUMS (verify with `sha256sum -c`).